

## **Subject Description Form**

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| <b>Subject Code</b>                                          | ITC3202T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Subject Title</b>                                         | <b>Woven Fabric Technology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Credit Value</b>                                          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Level</b>                                                 | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pre-requisite(s) / &lt;Co-requisite&gt; / (Exclusion)</b> | ITC2202T Foundations of Textiles I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Objectives</b>                                            | The subject provides a more comprehensive knowledge in fabric technology as required by a textile technologist.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Intended Learning Outcomes</b>                            | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>describe and explain the principle of woven fabric formation;</li> <li>compare different shedding systems, different weft yarn insertion systems and their applications in woven fabric production;</li> <li>identify and compare the structures and characteristics of different types of woven fabrics;</li> <li>identify different types of non-woven technologies from their processing features, types of products and applications of such products;</li> <li>communicate efficiently and clearly with textile professionals.</li> </ol> |
| <b>Subject Synopsis/ Indicative Syllabus</b>                 | <p><b>(I) Woven Fabrics</b><br/>Weaving preparation and weaving process. Different shedding systems and different weft yarn insertion systems of weaving machines. Derivatives of elementary weaves. Combined weaves. Color and weave effect. Jacquard fabrics. Specialized fabrics. Tri-axial woven fabrics. Three-dimensional woven fabrics and woven fabric construction.</p> <p><b>(II) Non-woven Fabric Technologies</b><br/>Web forming methods (dry-laid, wet-laid and direct-laid) Web bonding methods (mechanical bonding, thermal bonding and chemical bonding). Non-woven fabric structures, properties and applications.</p>               |
| <b>Teaching/Learning Methodology</b>                         | Teaching will be primarily conducted in lectures. Appropriate demonstration of fabric samples, their production machineries and related tools and instruments will be conducted in the laboratories.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| Assessment Methods in Alignment with Intended Learning Outcomes                                           | Specific assessment methods/tasks                                                                                                                                                                                                                                                                                                                   | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |   |          |   |
|                                                                                                           |                                                                                                                                                                                                                                                                                                                                                     |             | a                                                                              | b | c | d        | e |
|                                                                                                           | Continuous assessment                                                                                                                                                                                                                                                                                                                               | 50%         | ✓                                                                              | ✓ | ✓ | ✓        | ✓ |
|                                                                                                           | 1. Test                                                                                                                                                                                                                                                                                                                                             | 25%         | ✓                                                                              | ✓ | ✓ |          |   |
|                                                                                                           | 2. Report                                                                                                                                                                                                                                                                                                                                           | 25%         |                                                                                | ✓ | ✓ | ✓        | ✓ |
|                                                                                                           | Examination                                                                                                                                                                                                                                                                                                                                         | 50%         | ✓                                                                              | ✓ | ✓ | ✓        | ✓ |
|                                                                                                           | Total                                                                                                                                                                                                                                                                                                                                               | 100%        |                                                                                |   |   |          |   |
| Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: | This subject will provide fabric knowledge and laboratory experience to students for understanding fabric technology. The coursework will provide a continuing assessment of the student’s learning outcomes at different stages and aspects of the subject, and a final examination will assess the overall understanding of the subject contents. |             |                                                                                |   |   |          |   |
|                                                                                                           |                                                                                                                                                                                                                                                                                                                                                     |             |                                                                                |   |   |          |   |
| Student Study Effort Expected                                                                             | Class contact:                                                                                                                                                                                                                                                                                                                                      |             |                                                                                |   |   |          |   |
|                                                                                                           | • Lecture                                                                                                                                                                                                                                                                                                                                           |             |                                                                                |   |   | 21 Hrs.  |   |
|                                                                                                           | • Lab                                                                                                                                                                                                                                                                                                                                               |             |                                                                                |   |   | 18 Hrs.  |   |
|                                                                                                           | Other student study effort:                                                                                                                                                                                                                                                                                                                         |             |                                                                                |   |   |          |   |
|                                                                                                           | • Self-study                                                                                                                                                                                                                                                                                                                                        |             |                                                                                |   |   | 66 Hrs.  |   |
|                                                                                                           | Total student study effort                                                                                                                                                                                                                                                                                                                          |             |                                                                                |   |   | 105 Hrs. |   |
| Reading List and References                                                                               | <b>Essential reading</b>                                                                                                                                                                                                                                                                                                                            |             |                                                                                |   |   |          |   |
|                                                                                                           | Adanur, S. (2001), <i>Handbook of Weaving</i> . Technomic Pub. Co. Inc.                                                                                                                                                                                                                                                                             |             |                                                                                |   |   |          |   |
|                                                                                                           | Grosiki, Z. J. (1975), <i>Watson’s Textile Design &amp; Color</i> . 7th ed., Newnes-Butterworths.                                                                                                                                                                                                                                                   |             |                                                                                |   |   |          |   |
|                                                                                                           | Grosiki, Z. J. (1977), <i>Watson’s Advanced Textile Design &amp; Compound Woven Structures</i> . 4th ed., Newnes-Butterworths.                                                                                                                                                                                                                      |             |                                                                                |   |   |          |   |
|                                                                                                           | Hu, J. L. (2008), <i>3-D Fibrous Assemblies</i> . Woodhead Pub; CRC Press.                                                                                                                                                                                                                                                                          |             |                                                                                |   |   |          |   |
|                                                                                                           | Russell, S. J. (2007), <i>Handbook of Nonwovens</i> . Woodhead Pub; CRC Press.                                                                                                                                                                                                                                                                      |             |                                                                                |   |   |          |   |

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|  | <p><b><u>Supplementary</u></b></p> <p>Jiang, S. Q. and Hua, T. (2013), <i>Digital Jacquard: Mythologies</i>. The Hong Kong Polytechnic University.</p> <p>Shaikh, I. A. (2005), <i>Pocket Weaving Expert</i>. Textile training books, Textile InfoSociety.</p> |
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